

Batch: C3 Roll No.: 16010123325**Experiment No. 6****Grade: AA / AB / BB / BC / CC / CD / DD****Title:** Implementing Database as a Service on Azure Cloud and performing CRUD operations on the database.**Objective:**

To create and configure a Database as a Service (DBaaS) on the Microsoft Azure cloud platform and perform CRUD (Create, Read, Update, Delete) operations on the database using a client application or query interface.

Expected Outcome of Experiment:

CO	Outcome
CO4	Build cloud services and applications

Books/ Journals/ Websites referred:

<https://learn.microsoft.com/en-us/azure/azure-sql/database/>
<https://learn.microsoft.com/en-us/azure/azure-sql/database/single-database-create-quickstart?view=azuresql>

Related Theory: -

Database as a Service (DBaaS) is a cloud computing service model in which the cloud provider offers fully managed database services over the internet. Users can create, configure, and use databases without worrying about hardware provisioning, operating systems, or database server maintenance. In this service the cloud provider manages infrastructure, OS, database engine installation, patching, backups, and availability. The consumer focuses only on data, schema design, and queries.

Key Components of DBaaS

1. **Database Engine**
 - Azure SQL Database (relational DBMS)
2. **Storage Management**
 - Automatic data storage and scaling
 - Built-in backup and recovery
3. **Networking**
 - Secure connectivity via firewall rules
 - Public/private access configuration
4. **Management Interface**
 - Azure Portal
 - SQL Query Editor
 - Client tools (SSMS)
5. **Security & Availability**
 - Authentication & authorization
 - High availability and fault tolerance

Responsibility Division in DBaaS:

Layer	Managed By
Physical Data Center	Cloud Provider
Servers & Storage	Cloud Provider
Operating System	Cloud Provider
Database Engine	Cloud Provider
Backup & Patching	Cloud Provider
Database Schema & Data	Consumer
Queries & Applications	Consumer

Characteristics of DBaaS

1. **On-Demand Provisioning**
 Databases can be created instantly through the Azure portal without manual setup.

2. Scalability
Database storage and performance can be scaled up or down based on workload.
3. Managed Maintenance
Patching, updates, backups, and monitoring are handled automatically.
4. High Availability
DBaaS ensures minimal downtime using replication and fault-tolerant architecture.
5. Pay-as-You-Use
Users are billed only for the database resources they consume.

Implementation Details

Follow the steps provided in the reference link or Azure documentation:

1. Create an **Azure SQL Database** using the Azure Portal.
2. Configure database server settings (authentication, firewall rules).
3. Connect to the database using:
 - Azure Query Editor or
 - SQL Server Management Studio (SSMS).
4. Create a sample table in the database.
5. Perform **CRUD operations**:
 - **Create** – Insert records
 - **Read** – Retrieve records
 - **Update** – Modify existing records
 - **Delete** – Remove records

Sample CRUD Operations:

1. -- Create

```
INSERT INTO Student VALUES (1, 'Amit', 21);
```

2. -- Read

```
SELECT * FROM Student;
```

3. Update

UPDATE Student SET Age = 22 WHERE ID = 1;

4. Delete

DELETE FROM Student WHERE ID = 1;

Create an Azure SQL Database

Home > Azure SQL | SQL databases

Create SQL Database

Microsoft

×

Basics

Networking

Security

Additional settings

Tags

Review + create

Create a SQL database with your preferred configurations. Complete the Basics tab then go to Review + Create to provision with smart defaults, or visit each tab to customize. [Learn more](#)

✓ Create a serverless Azure SQL Database with 100,000 vCore seconds, 32GB of data, and 32GB of backup storage free per month for the lifetime of your subscription. Limit is 10 free databases per subscription. [Learn more](#)

Apply offer

✓ SQL Database Hyperscale: Low price, high scalability, and best feature set. [Learn more](#)

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Azure for Students

Resource group *

Select a resource group

[Create new](#)

Database details

Review + create

Next: Networking >

Server Set Up

Home > Azure SQL | SQL databases > Create SQL Database

Create SQL Database Server

Microsoft

Server name *

server-exp-6

.database.windows.net

Location *

(US) East US

Authentication

ℹ Azure Active Directory (Azure AD) is now Microsoft Entra ID. [Learn more](#)

Select your preferred authentication methods for accessing this server. We recommend using only Microsoft Entra authentication. [Learn more](#) using an existing Microsoft Entra user, group, or application as Microsoft Entra admin. [Learn more](#)

Authentication method

- ☐ Use Microsoft Entra-only authentication
☐ Use both SQL and Microsoft Entra authentication
☒ Use SQL authentication

Server admin login *

admin_shreya

Password *

Confirm password *

✓ Password and confirm password must match.

OK

Hardware & Service Tier

Home > Azure SQL | SQL databases

Configure

 Feedback

Service and compute tier

Select from the available tiers based on the needs of your workload. The vCore model provides a wide range of configuration controls and offers Hyperscale and Serverless to automatically scale your database based on your workload needs. Alternately, the DTU model provides set price/performance packages to choose from for easy configuration. [Learn more](#)

☒ SQL Database Hyperscale: Low price, high scalability, and best feature set. [Learn more](#)

Service tier: General Purpose (Most budget friendly) [Compare service tiers](#)

Compute tier:
☒ **Provisioned** - Compute resources are pre-allocated. Billed per hour based on vCores configured.
☐ **Serverless** - Compute resources are auto-scaled. Billed per second based on vCores used.

Compute Hardware

Select the hardware configuration based on your workload requirements. Availability of compute optimized, memory optimized, and confidential computing hardware depends on the region, service tier, and compute tier.

Hardware Configuration: **Standard-series (Gen5)**
 up to 128 vCores, up to 625 GB memory
[Change configuration](#)

[Apply](#)



Cost summary

General Purpose (GP, Gen5, 2)	
Cost per vCore (in USD)	195.21
vCores selected	x 2
Cost per GB (in USD)	0.13
Max storage selected (in GB)	x 41.6
ESTIMATED COST / MONTH	395.87 USD

Security Set Up

Home > Azure SQL | SQL databases

Create SQL Database

Microsoft

Basics Networking **Security** Additional settings Tags Review + create

Microsoft Defender for SQL

Protect your data using Microsoft Defender for SQL, a unified security package including vulnerability assessment and advanced threat protection for your server. [Learn more](#)

Get started with a 30 day free trial period, and then 15 USD/server/month.

Enable Microsoft Defender for SQL * ☐ Start free trial
☒ **Not now**

Ledger

Ledger cryptographically verifies the integrity of your data and detects any tampering that might have occurred. [Learn more](#)

Ledger: **Not configured**
[Configure ledger](#)

Server identity

Use system assigned and user assigned managed identities to enable central access management between this database and other Azure resources. [Learn more](#)

Identity settings for the existing server can be updated by navigating to 'server-exp-6'

[Review + create](#)

[< Previous](#)

[Next: Additional settings >](#)



Cost summary

General Purpose (GP, 5, Gen5, 1)	
Cost per GB (in USD)	0.13
Max storage selected (in GB)	x 41.6
ESTIMATED STORAGE COST / MONTH	5.45 USD
COMPUTE COST / VCORE SECOND¹	0.000159 USD

NOTES

¹ Serverless databases are billed in vCore seconds based on a combination of CPU and memory utilization. [Learn more about serverless billing](#)

Load DB with sample data loaded

Home > Azure SQL | SQL databases

Create SQL Database

Microsoft

Basics Networking Security **Additional settings** Tags Review + create

Customize additional configuration parameters including collation & sample data.

Data source

Start with a blank database, restore from a backup or select sample data to populate your new database.

Use existing data *

None Backup **Sample**

AdventureWorksLT will be created as the sample database.

Database collation

Database collation defines the rules that sort and compare data, and cannot be changed after database creation. The default database collation is SQL_Latin1_General_CP1_CI_AS. [Learn more](#)

Collation

Cost summary

General Purpose (GP_S, Gen5, 1)
Cost per GB (in USD) **0.13**
Max storage selected (in GB) **x 41.6**

ESTIMATED STORAGE COST / MONTH **5.45 USD**
COMPUTE COST / VCORE SECOND **0.000159 USD**

NOTES
* Serverless databases are billed in vCore seconds based on a combination of CPU and memory utilization. [Learn more about serverless billing](#)

Review + create < Previous Next : Tags >

Summary of DB Config + Create

Home > Azure SQL | SQL databases

Create SQL Database

Microsoft

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Terms

By clicking "Create", I (a) agree to the legal terms and privacy statement(s) associated with the Marketplace offering(s) listed above; (b) authorize Microsoft to bill my current payment method for the fees associated with the offering(s), with the same billing frequency as my Azure subscription; and (c) agree that Microsoft may share my contact, usage and transactional information with the provider(s) of the offering(s) for support, billing and other transactional activities. Microsoft does not provide rights for third-party offerings. For additional details see [Azure Marketplace Terms](#)

Basics

Subscription	Azure for Students
Resource group	exp6
Region	centralindia
Database name	wow
Server	server-exp-6
Compute + storage	General Purpose - Serverless: Standard-series (Gen5), 1 vCore, 32 GB storage, zone redundant disabled
Backup storage redundancy	Geo-redundant backup storage

Networking

Allow Azure services and resources to access this server	No
--	----

Cost summary

General Purpose (GP_S, Gen5, 1)
Cost per GB (in USD) **0.13**
Max storage selected (in GB) **x 41.6**

ESTIMATED STORAGE COST / MONTH **5.45 USD**
COMPUTE COST / VCORE SECOND **0.000159 USD**

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Create < Previous [Download a template for automation](#)

Run Query on the Sample DB

Home > exp6 > wow (server-exp-6/wow)

wow (server-exp-6/wow) | Query editor (preview) ☆ ...

SQL database

Search < Login + New Query Open query Feedback Getting started

wow (shreyamenon479@gmail.com)

Overview
Activity log
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Diagnose and solve problems
Query editor (preview)
Mirror database in Fabric (preview)
Resource visualizer
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Tables
Views
Stored Procedures

Query 1

Run Cancel query Save query Export data as Show only Editor

```
1 SELECT TOP 20 pc.Name AS CategoryName, p.Name AS ProductName
2 FROM SalesLT.ProductCategory pc JOIN SalesLT.Product p ON pc.ProductCategoryID = p.ProductCategoryID;
```

Results Messages

Search to filter items...

CategoryName	ProductName
Road Frames	HL Road Frame - Black, 58
Road Frames	HL Road Frame - Red, 58
Helmets	Sport-100 Helmet, Red
Helmets	Sport-100 Helmet, Black
Socks	Mountain Bike Socks, M

Query succeeded | 0s

CRUD on DB

Create Table

Home > exp6 > wow (server-exp-6/wow)

wow (server-exp-6/wow) | Query editor (preview) ☆ ...

SQL database

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Query 1

Run Cancel query Save query Export data as Show only Editor

```
1 CREATE TABLE F1 (
2   driver_id INT PRIMARY KEY,
3   driver_name VARCHAR(50),
4   team VARCHAR(50),
5   country VARCHAR(50),
6   championships INT
7 );
```

Results Messages

Query succeeded: Affected rows: 0

Query succeeded | 0s

Insert Values into Table

Home > exp6 > wow (server-exp-6/wow)

wow (server-exp-6/wow) | Query editor (preview) ☆ ...

SQL database

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wow (shreyamenon479@gmail.com)

Tables
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Query 1 ×

Run Cancel query Save query Export data as Show only Editor

```
1 INSERT INTO F1 VALUES (2, 'Lewis Hamilton', 'Mercedes', 'UK', 7);
```

Results Messages

Query succeeded: Affected rows: 1

Query succeeded | 0s

Display Table

Home > exp6 > wow (server-exp-6/wow)

wow (server-exp-6/wow) | Query editor (preview) ☆ ...

SQL database

Search Login + New Query Open query Feedback Getting started

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wow (shreyamenon479@gmail.com)

Tables
Views
Stored Procedures

Query 1 ×

Run Cancel query Save query Export data as Show only Editor

```
1 SELECT driver_name, team FROM F1;
```

Results Messages

Search to filter items...

driver_name	team
Lewis Hamilton	Mercedes

Query succeeded | 0s

Update DB content

Home > exp6 > wow (server-exp-6/wow)

wow (server-exp-6/wow) | Query editor (preview) ☆ ...

SQL database

Search

Login + New Query Open query Feedback Getting started

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wow (shreyamenon479@gmail.com)

Tables Views Stored Procedures

Query 1

Run Cancel query Save query Export data as Show only Editor

```
1 UPDATE F1
2 SET team = 'Ferrari'
3 WHERE driver_id = 2;
```

Results Messages

Query succeeded: Affected rows: 1

Query succeeded | 0s

Delete DB

Home > exp6 > wow (server-exp-6/wow)

wow (server-exp-6/wow) | Query editor (preview) ☆ ...

SQL database

Search

Login + New Query Open query Feedback Getting started

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wow (shreyamenon479@gmail.com)

Tables Views Stored Procedures

Query 1

Run Cancel query Save query Export data as Show only Editor

```
1 DELETE FROM F1
2 WHERE driver_id = 2;
```

Results Messages

Query succeeded: Affected rows: 1

Query succeeded | 0s

Post Lab Questions:

Q1. How does Database as a Service (DBaaS) reduce administrative overhead compared to managing an on-premise database server?

Department of Computer Engineering

Database as a Service (DBaaS) significantly reduces administrative overhead by shifting database management responsibilities from the user to the cloud service provider. Key ways this happens are:

- **No infrastructure management:** Hardware provisioning, server setup, and data center maintenance are handled by the provider.
- **Automated installation & updates:** Database software installation, patching, and version upgrades are performed automatically.
- **Simplified scaling:** Storage and compute resources can be scaled up or down on demand without manual intervention.
- **Reduced staffing needs:** Organizations don't need dedicated DBAs for routine tasks like monitoring, tuning, and maintenance.
- **Pay-as-you-use model:** Eliminates costs and effort associated with capacity planning and idle resources.

Overall, DBaaS allows developers and organizations to focus more on **application development and data usage**, rather than database administration.

Q2. Why is DBaaS preferred for applications requiring high availability and automatic backup?

DBaaS is preferred for such applications because it provides **built-in reliability, fault tolerance, and data protection** features that are difficult and costly to implement on-premise:

- **High availability by design:** DBaaS platforms use replication, clustering, and multi-zone deployments to ensure minimal downtime.
- **Automatic backups:** Databases are backed up at regular intervals without manual configuration, reducing the risk of data loss.
- **Fast recovery:** Point-in-time recovery and automated restore options help quickly recover from failures or accidental data deletion.
- **Disaster recovery support:** Data is often replicated across geographically distributed locations.
- **SLA-backed uptime guarantees:** Cloud providers offer service-level agreements ensuring consistent availability.

As a result, DBaaS is ideal for **mission-critical applications** where continuous access to data and data safety are essential.

Conclusion:-

In this experiment, a Database as a Service (DBaaS) was successfully created and configured on the Microsoft Azure cloud platform. The database was deployed without the need to manage physical hardware, operating systems, or database server installation. CRUD (Create, Read, Update, Delete)

Department of Computer Engineering

operations were performed using a query interface, demonstrating how cloud-managed databases simplify development and maintenance tasks.